a. Blood lake in Texas-the blood red colour is due to the excess presence of purple sulphur bacteria.

b. Giant tube worms seen in deep sea hydrothermal vents survive on nutrients given by chemolithotrophic bacteria.



a)



b)

6.4 Photosynthesis

Photosynthesis is a process of capturing light energy and converting into chemical energy. The chemical energy produced in the form of ATP and NADPH is used to synthesise organic compounds (carbohydrates); to be used as food. This ability makes photosynthesis, a significant process taking place on earth.

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Eukaryotes (plants and algae) and prokaryotes (cyanobacteria and purple, green bacteria) are capable of carrying out photosynthesis. Cyanobacteria perform photosynthesis in a similar manner to plants.

Photosynthesis in Bacteria

There are four groups of photosynthetic bacteria. They are green sulphur bacteria

Table 6.2: Nutritional classes of Microorganisms

Nutritional class	Energy/Electron/Carbon source	Organisms
Litho photoautotrophs	Light energy Inorganic e^- donor CO_2	Cyanobacteria, Purple and Green sulphur Bacteria
Organo photoheterotrophs	Light energy Organic e^- donor Organic carbon source	Purple and Green Nonsulfur bacteria
Ltiho chemoautotrophs	Inorganic chemical compounds as energy source Inorganic e^- donor CO_2	Nitrifying bacteria, Iron bacteria
Organo chemoheterotrophs	Organic compounds as energy, electron and carbon source.	Most pathogenic bacteria, fungi and protozoa.

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